

The empirical recurrence for OEIS A230678, recovered from its tail

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8 September 2026

Abstract

OEIS A230678 records “Empirical recurrence of order 64” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 78$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 66$. Its last coefficient is $c_{64} = 2523136 \neq 0$, so no further tail satisfies anything shorter; any order-64 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A230678 is “Number of $4 \times n$ 0..2 arrays $x(i,j)$ with each element horizontally, diagonally or antidiagonally next to at least one element with value $2 - x(i,j)$.”. It has offset 1 and begins

0, 423, 17271, 730503, 37034355, 1844154933, 91258562955, 4522036938501, ...

The entry states:

Empirical recurrence of order 64 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 08:45:32, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 78$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 78$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 64: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 64.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 156$ exact terms from $n = 2$ on, it returns order 64 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{64} c_i a(n-i),$$

c_1	33	c_{17}	−725358827070	c_{33}	216769426062920	c_{49}	387147770768
c_2	629	c_{18}	−971402562457	c_{34}	188939845514619	c_{50}	560132773804
c_3	8527	c_{19}	2519469403594	c_{35}	−13118312311714	c_{51}	151726557152
c_4	50224	c_{20}	10180635391581	c_{36}	−14597736584455	c_{52}	−546954180686
c_5	−41280	c_{21}	10118590722792	c_{37}	77147515911126	c_{53}	−114133959294
c_6	−3950194	c_{22}	−6176450025631	c_{38}	31033233134298	c_{54}	−2744555600
c_7	−29712481	c_{23}	−17721896890756	c_{39}	−10776275747989	c_{55}	76236025080
c_8	−139814523	c_{24}	−38922915329665	c_{40}	−19531183709989	c_{56}	−19759006800
c_9	−176096112	c_{25}	−29312475658925	c_{41}	19695276022691	c_{57}	−20045094528
c_{10}	686849163	c_{26}	60393568679467	c_{42}	−8728319751305	c_{58}	−10661881536
c_{11}	4893067446	c_{27}	74244547534692	c_{43}	−54897696905580	c_{59}	−748342528
c_{12}	12077099064	c_{28}	4426531798749	c_{44}	−34210789878166	c_{60}	2450107392
c_{13}	−7319520683	c_{29}	−83970020750493	c_{45}	4326706535614	c_{61}	1070550016
c_{14}	3910878837	c_{30}	−214724469616017	c_{46}	17457899776451	c_{62}	104726528
c_{15}	−58231207283	c_{31}	−182202260337705	c_{47}	7937489383480	c_{63}	5169152
c_{16}	−418719978877	c_{32}	23298657459603	c_{48}	−521094951961	c_{64}	2523136

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 66$, and it is the recurrence OEIS A230678 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{64} - c_1 x^{64-1} - \dots - c_{64}$. Its constant term is $-c_{64} = -2523136$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 64 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 64, so $\deg q = 64$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 64, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 64, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 2523136.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-64 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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